

Ayush Badwaik - CM23051

Machine Vision – Prelab

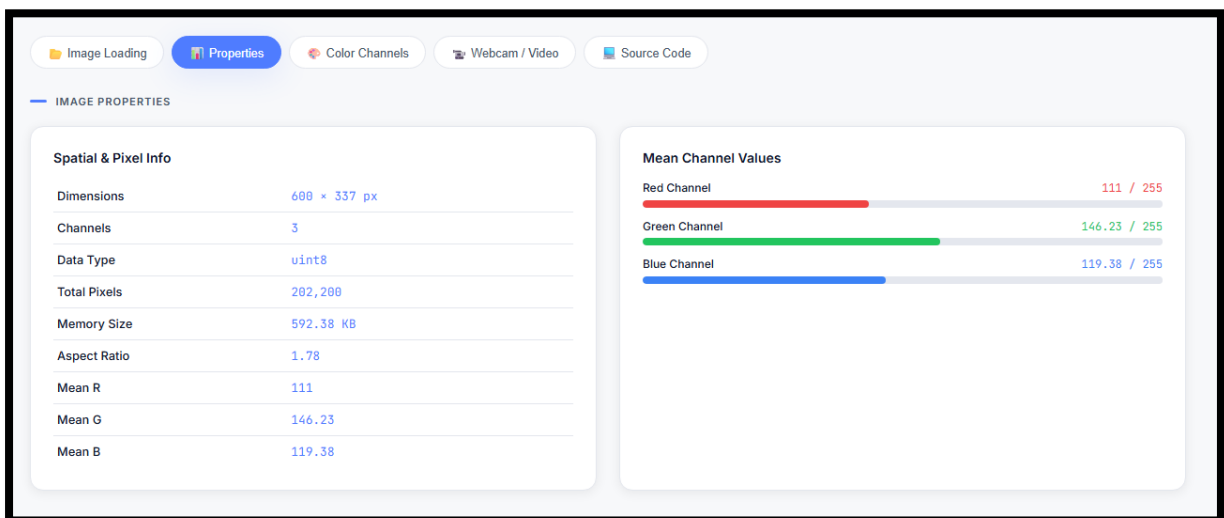
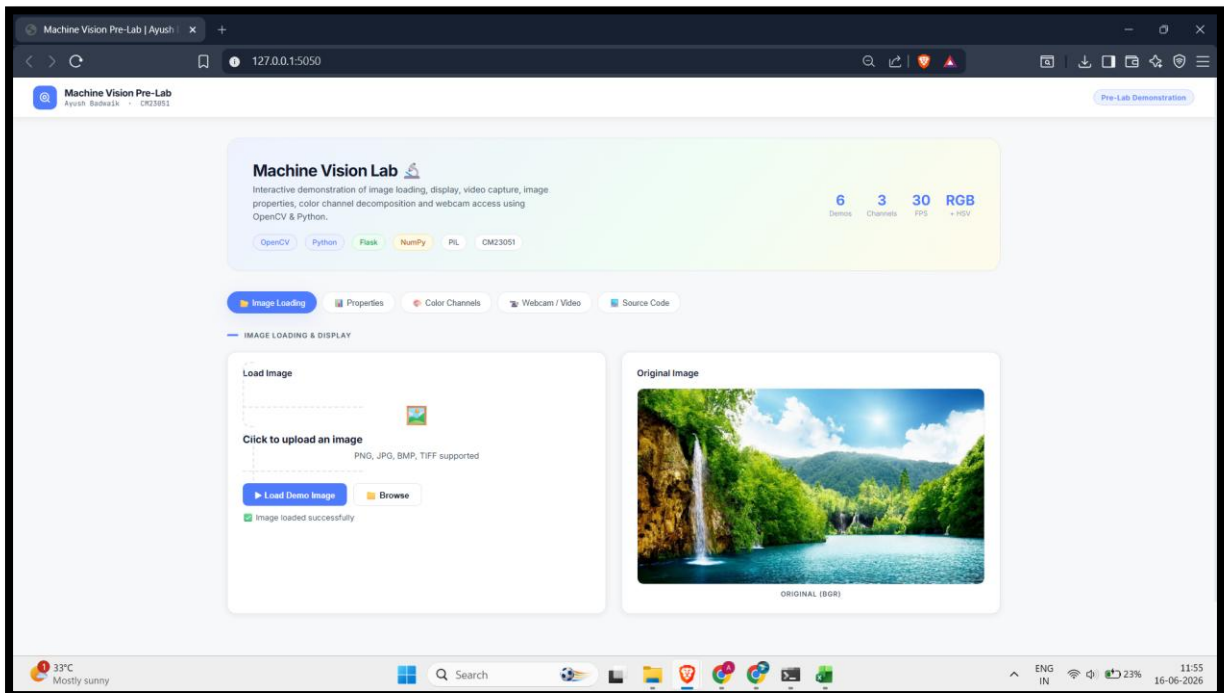


Image Loading

Properties

Color Channels

Webcam / Video

Source Code

COLOR CHANNEL DECOMPOSITION

Channel Visualisation



ORIGINAL



RED CHANNEL



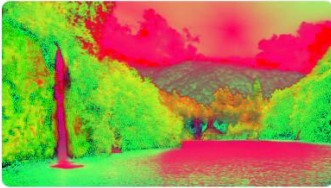
GREEN CHANNEL



BLUE CHANNEL



GRAYSCALE



HSV

Machine Vision Pre-Lab · Ayush Badwalk · CM23051 · OpenCV | Python | Flask | NumPy

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WEBCAM & VIDEO CAPTURE

Live Video Feed

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11:55:43

● Live — capturing frames at 30 fps

▶ Start Webcam

■ Stop

📷 Snapshot

Captured Frame Analysis



Width × Height

640 × 480

Channels

3

Size

900 KB

Mean R/G/B

148.87 / 144.22 / 143.09

[Image Loading](#) [Properties](#) [Color Channels](#) [Webcam / Video](#) [Source Code](#)

SOURCE CODE REFERENCE

Image Loading & Properties (OpenCV)

```
# 1. Image Loading: Import cv2, numpy as np, img =
cv2.imread("image.jpg") # img is a numpy array of 8 x 8 x 3 in BGR order
# 2. Image Properties: h, w, c = img.shape # height, width, channels
dtype = img.dtype # uint8 nbytes = img.nbytes # total bytes in memory #
# Displaying with imshow cv2.imshow("Window", img) cv2.waitKey(0)
cv2.destroyAllWindows()
```

Color Channel Splitting

```
# Split into B, G, R channels: b, g, r = cv2.split(img) # Reconstruct
individual channel: blue zero = np.zeros_like(b) r_vis =
cv2.merge([zero, zero, r]) g_vis = cv2.merge([zero, g, zero]) b_vis =
cv2.merge([b, zero, zero]) # Convert to grayscale: gray =
cv2.cvtColor(img, cv2.COLOR_BGR2GRAY) # Convert to RGB: rgb =
cv2.cvtColor(img, cv2.COLOR_BGR2RGB) # Channel mean values: mean =
cv2.mean(img) # B, G, R
```

Webcam / Video Capture

```
# Access webcam: import cv2, capture camera: cap = cv2.VideoCapture(0)
while cap.isOpened(): ret, frame = cap.read() if not ret: break #
# Process frame: gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
cv2.imshow("Webcam", frame) if cv2.waitKey(1) & 0xFF == ord('q'): break
cap.release() cv2.destroyAllWindows()
```

Video File Capture & Saving

```
# Read from video file: cap = cv2.VideoCapture("video.mp4") fps =
cap.get(cv2.CAP_PROP_FPS) w = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH)) h
= int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT)) # VideoWriter to save output
fourcc = cv2.VideoWriter_fourcc(*'mp4v') out =
cv2.VideoWriter('out.mp4', fourcc, fps, (w, h)) while cap.isOpened():
ret, frame = cap.read() if not ret: break out.write(frame)
cap.release(); out.release()
```